

DAVID M. VANCE

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Location: Austin, TX

PRINCIPAL TECHNICAL ARCHITECT & ENTERPRISE STRATEGIST

PROFESSIONAL SUMMARY

Visionary and highly accomplished Principal Technical Architect with over 25 years of progressive experience leading enterprise software architecture, global infrastructure engineering, and digital transformation initiatives. Expertise across large-scale distributed systems, multi-cloud computing paradigms, cloud-native modernization, and big data governance architectures. Proven track record of guiding fortune-500 financial institutions, global e-commerce systems, and high-growth SaaS corporations through critical technological evolutions. Highly skilled in aligning complex technical roadmaps with business objectives, managing multi-million dollar infrastructure budgets, and steering top-performing, cross-functional engineering matrix groups across the globe. Recognized authority in building resilient, highly secure, hyper-scalable systems capable of handling billions of transactions daily while maintaining stringent 99.999% high-availability operational baselines.

TECHNICAL ECOSYSTEM MATRIX

Domain / Category	Technologies & Frameworks Proficiencies
Cloud Computing & Infra	Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure, OpenStack, Terraform, CloudFormation, Ansible, Pulumi
Architecture Styles	Microservices, Event-Driven Architecture (EDA), Domain-Driven Design (DDD), Serverless, Service Mesh (Istio), GraphQL, REST, gRPC
Containerization / Orchestration	Kubernetes (EKS, GKE, AKS), Docker, OpenShift, Nomad, Helm, Kustomize, Rancher
Data Management & Stores	Apache Kafka, Apache Cassandra, PostgreSQL, MongoDB, Redis, AWS DynamoDB, Oracle RAC, Snowflake, Neo4j, Couchbase
Programming Languages	Java / Spring Boot, Go, Python, Scala, C++, Rust, TypeScript, Bash, SQL
CI/CD & DevOps Pipelines	Jenkins, GitLab CI, GitHub Actions, Spinnaker, ArgoCD, SonarQube, Artifactory, Tekton
Security & Compliance	OAuth 2.0, OIDC, SAML, HashiCorp Vault, TLS/mTLS, AWS IAM, WAF, Zero-Trust Architecture, GDPR, HIPAA, PCI-DSS compliance
Observability & Telemetry	Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), Datadog, OpenTelemetry, Jaeger, Splunk, Dynatrace

PROFESSIONAL EXPERIENCE

NEXUS GLOBAL CLOUD SYSTEMS | Austin, TX

Principal Technical Architect (January 2021 – Present)

Architectural lead for a premier global cloud platform provider, spearheading the core platform modernization strategy for 500+ enterprise microservices hosting billions of API connections.

- Directed the transition of a legacy bare-metal infrastructure to a multi-region distributed hybrid cloud platform using AWS and GCP, improving scalability while dropping capital operational expenditures by \$14M annually.
- Engineered an enterprise-wide event mesh framework utilizing Apache Kafka and AWS Kinesis that successfully processes over 15 Billion transactions per day with a median P99 edge-latency of less than 12 milliseconds.
- Pioneered a unified declarative governance system using HashiCorp Terraform and GitOps practices (ArgoCD), reducing infrastructure deployment lifecycles from 3 weeks to under 8 minutes with fully embedded automated compliance tests.
- Spearheaded an executive technical strategy council that consolidated architectural oversight across 6 disparate global business pillars, resolving multi-year technical barriers and accelerating platform feature delivery speeds by 35%.
- Established comprehensive Zero-Trust network topologies utilizing Istio service mesh, mTLS authentication models, and automated SPIFFE/SPIRE cryptographic workload provisioning across 40,000+ container endpoints.
- Mentored and directly upskilled 12 Staff and Lead Architects, fostering an engineering culture that achieved 98% retention over 5 consecutive quarters.

VANGUARD FINANCIAL TECHNOLOGIES | New York, NY

Senior Enterprise Architect (March 2018 – December 2020)

Promoted to enterprise-wide architecture team to design and standardize core banking transaction frameworks across international retail branches.

- Architected a ledger processing framework using Java, Spring Boot, and Apache Cassandra that supported transactional workloads peaking at 120,000 requests per second under stringent PCI-DSS Level 1 conditions.
- Created a real-time predictive fraud-detection system pipeline utilizing Apache Flink and Redis, allowing transaction risk evaluations to execute inline within a strict 45ms window, saving \$8.5M in fraud losses.
- Managed a technical infrastructure review board governing an annual operational budget of \$28M, aligning cloud spending optimizations that eliminated under-utilized capacities.

Lead Solutions Architect (September 2015 – March 2018)

Initially recruited to modernize legacy mainframe payment gateways into scalable microservice cloud systems.

- Oversaw the migration of 4 main clearinghouse system frameworks into a cloud-native microservices mesh deployed on AWS EKS, resulting in an immediate 60% reduction in processing window delays.
- Architected fallback data patterns across decoupled hybrid nodes to guarantee continuous transaction routing through unpredictable distributed network disconnects, achieving 99.999% uptime metrics over 3 consecutive years.
- Authored 45+ comprehensive enterprise architecture reference blueprints covering identity propagation, global database synchronization, and automated disaster recovery execution patterns.

E-COMMERCE UNIFIED INC. | Seattle, WA

Principal Software Architect (November 2011 – August 2015)

Served as chief technical strategist for a global B2B digital commerce platform handling \$18B in annualized gross merchandise value.

- Designed a modern distributed catalog search engine leveraging Elasticsearch and distributed Redis clusters, accelerating item search rendering speeds by 300% during seasonal high-traffic events.
- Engineered an asynchronous inventory allocation model using Go and RabbitMQ that eliminated transaction race-conditions across multi-node distribution warehouses, cutting order cancellation frequencies by 18%.
- Pioneered a multi-tenant SaaS integration layer facilitating localized multi-currency and tax compliance calculation adapters across 34 distinct international zones.
- Orchestrated standard software delivery automation transformations utilizing Jenkins distributed pipeline grids, shrinking production release cadences from bi-monthly schedules to multiple zero-downtime daily deployments.

APEX STREAMING & MEDIA SOLUTIONS | Los Gatos, CA

Senior Systems Architect (August 2008 – October 2011)

Designed back-end ingestion, distributed caching tiers, and content distribution networks (CDN) optimized for large-scale media file streaming profiles.

- Architected a global dynamic edge-routing software framework using C++ and Lua extensions, shifting 40% of static streaming workloads away from central cloud instances directly to regional network edge nodes.
- Structured data processing routines written in Python and C++ to normalize diverse incoming video file types into uniform multi-bitrate streams, optimizing storage foot-prints by 4.2 Petabytes.
- Coordinated direct joint technical integrations alongside external Tier-1 telecom carrier networks to map and resolve regional packet latency drops during widespread high-demand broadcast cycles.

ORION HEALTHCARE INFORMATICS | Boston, MA

Enterprise Solutions Architect (January 2005 – July 2008)

Led software system strategy for clinical health records processing systems across large healthcare provider ecosystems.

- Maintained complete technical ownership over a clinical record ingestion system processing 50M+ electronic updates daily, fully enforcing HIPAA and local privacy compliance regulations.
- Designed a secure cross-institutional tokenized data sharing exchange API utilizing SOAP/REST, Web Security extensions, and strong PKI infrastructure frameworks.
- Guided an 85-engineer matrix organization through technical transitions from archaic procedural code into object-oriented enterprise Java (J2EE) design patterns.

INTELLICOM NETWORKS SOLUTIONS | San Jose, CA

Senior Software Engineer / Tech Lead (June 2002 – December 2004)

Promoted to Lead role to spearhead software feature additions for high-throughput core network routing devices.

- Engineered multi-threaded packet inspection logic components written in C and C++, maximizing traffic throughput by 22% across embedded system limits.
- Created modular networking protocol stacks (TCP/IP, BGP extensions) following rigid real-time memory and processing utilization limits.
- Supervised a localized engineering pod of 8 developers, delivering high-priority firmware updates ahead of tight carrier hardware release timelines.

Software Engineer II (February 2000 – June 2002)

Responsible for developing, debugging, and maintaining foundational embedded operating system modules.

- Maintained and refactored device driver routines interacting with specialized physical hardware ASIC nodes.
- Developed a automated software profiling validation test harness, cutting regression bug escape tracking rates from 14% down to under 2%.
- Participated in core system code audits, optimizing critical memory block allocations to prevent historical leak failures.

GENESIS SOFTWARE CONSULTING | Dallas, TX

Junior Application Developer (July 1999 – January 2000)

Began professional technical career by contributing to multi-tier web database applications for regional commercial clients.

- Developed robust data access structures utilizing SQL Server, transactional procedures, and early web scripting technologies.
- Authored 120+ pages of detailed database schema descriptions and software component functional specifications.
- Resolved incoming Tier-3 software support bug queues, gaining deep insight into unexpected production environment edge cases.

REPRESENTATIVE ARCHITECTURAL PROJECTS PORTFOLIO

Project Context & Scope	Architectural Approach	Measurable Business Impact
Global Cloud Platform Modernization Nexus Global Cloud	Multi-region AWS/GCP hybrid network mesh topology using Infrastructure as Code (Terraform), Kubernetes orchestration, and Istio security wrappers.	Eliminated single-point infrastructure failures, yielding a 99.999% application uptime index and reducing yearly operational expenditures by \$14 Million.
Real-Time Event Processing Mesh Vanguard Financial Tech	Decoupled messaging architecture fueled by high-throughput Apache Kafka event topologies alongside inline Apache Flink streaming analytical data paths.	Scaled streaming handling capabilities to over 15 Billion tracking payloads daily while dropping system fraud exposure limits by \$8.5M.
Distributed E-Commerce Engine E-Commerce Unified Inc.	Asynchronous event-driven inventory allocation system written in Go, RabbitMQ message brokers, and horizontal Elasticsearch index clusters.	Accelerated catalog retrieval performance metrics by 300% and lowered holiday peak transaction dropouts by 18%.

KEY PROFESSIONAL ACHIEVEMENTS

- **Cloud Efficiency & Optimization Award (2023):** Recognized across Nexus Global Cloud for redesigning data ingress routes, saving \$4.2M in network egress costs.
- **Patents Filed (2019):** Co-inventor on Patent US-1049281-B2: 'Distributed Transaction Routing Protocol and Fallback Token Validation Across Hybrid Cloud Nodes.'
- **Corporate Open Source Excellence Leader (2016):** Spearheaded the internal engineering initiative to patch and contribute scalability features directly back to the upstream Apache Kafka open-source codebase.
- **Technical Keynote Speaker:** Invited domain panelist at international technology conventions including AWS re:Invent (2022) and CloudNativeCon/KubeCon Europe (2024).

PROFESSIONAL CERTIFICATIONS

- AWS Certified Solutions Architect – Professional (Recertified 2024)
 - HashiCorp Certified: Terraform Associate (2022)
 - TOGAF 9.2 Certified Enterprise Enterprise Architect (2016)
- Google Cloud Certified Professional Cloud Architect (2023)
 - Certified Kubernetes Administrator (CKA) – CNCF (2021)
 - Scaled Agile Framework (SAFe) 5 Product Owner/Product Manager (2019)

EDUCATION

Master of Science (M.S.) in Computer Science & Distributed Systems

The University of Texas at Austin | Graduated May 2002

Key Focus Area: High-Performance Computing, Cryptographic Network Protocols, and Advanced Database Systems Architecture.

Bachelor of Science (B.S.) in Computer Science

Texas A&M University, College Station, TX | Graduated May 1999

Honors: Magna Cum Laude | Tau Beta Pi Engineering Honor Society Member.